

Questions: The quotient rule

Sara Delgado Garcia

Summary

A selection of questions for the study guide on the quotient rule.

Before attempting these questions, it is highly recommended that you read [Guide: The quotient rule](#).

Differentiate the following functions using the quotient rule.

1.1. $\frac{e^x}{x}$

1.2. $\frac{e^{7x}}{x^5}$

1.3. $\frac{\ln(x)}{x^2}$

1.4. $\frac{e^{-x}}{x^2 + 11x - 2}$

1.5. $\frac{x^3 + 5x - 5}{x^2 + 3}$

1.6. $\frac{\cos(x)}{x^2 + 3x - 1}$

1.7. $\frac{\tan(x)}{\cos(x)}$

1.8. $\frac{\ln(3x)}{\ln(5) + x}$

1.9. $\frac{x^2 + 3x}{\cos(x)}$

1.10. $\frac{\ln(x)}{x^3 + 3}$.

1.11. $\frac{5 \tan(x)}{x}$.

1.12. $\frac{3x^7 - 27x^2 + 2\sqrt{x}}{x^2 + 1}$.

1.13. $\frac{e^{-3x}}{e^{2x}}$.

1.14. $\frac{e^3 x^3}{e^x}$.

1.15. $\frac{x^5}{x^5 + 1}.$

1.16. $\frac{\tan(x)}{\ln(x)}.$

1.17. $\frac{3 \sin(x)}{\ln(x)}$

1.18. $\frac{\tan(x) + 5x}{\sec(3x)}$

[After attempting the questions above, please click this link to find the answers.](#)

Version history and licensing

v1.0: initial version created 05/25 by Sara Delgado Garcia as part of a University of St Andrews VIP project.

[This work is licensed under CC BY-NC-SA 4.0.](#)